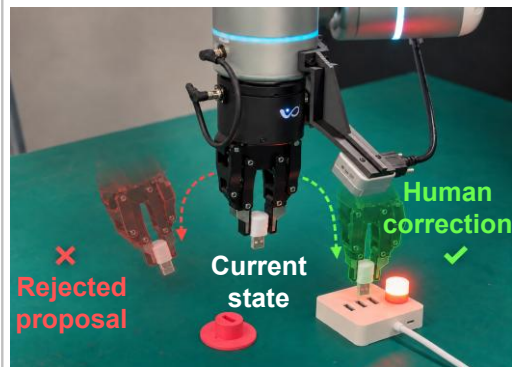
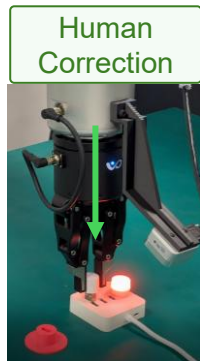
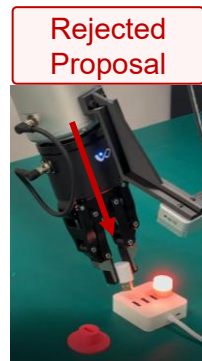


(a) Learn from Interventions



Learn Action Value

(b) Corrective Value Learning (CVL)

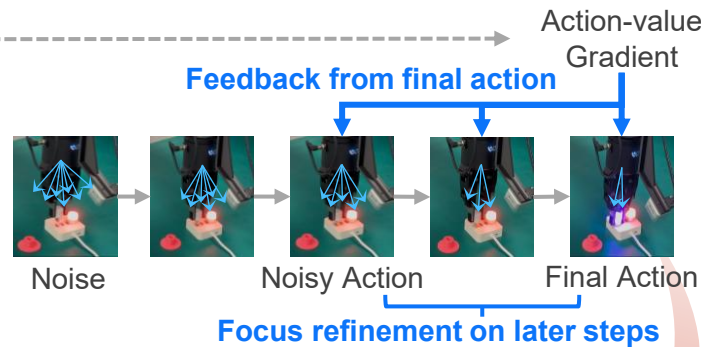


Guide VLA Improvement

Preference Ranking

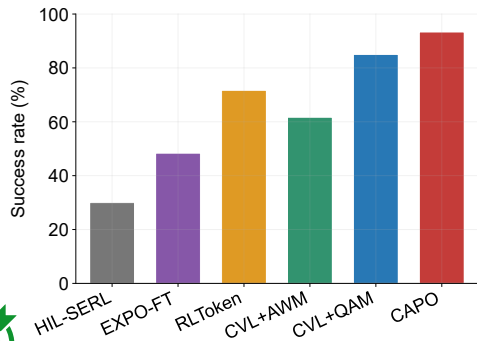


(c) Adjoint-based Policy Optimization (APO)

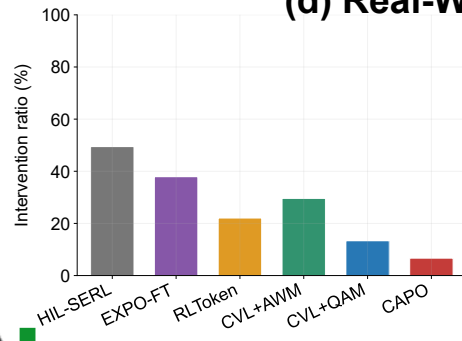


Execute Autonomously

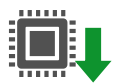
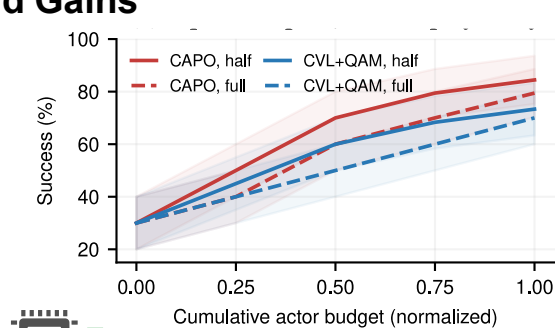
(d) Real-World Gains



Higher autonomous success



Fewer human interventions



Less backward computation



Precise Manipulation